International Journal of Infectious Diseases 70 (2018) 15–19

Contents lists available at ScienceDirect

International Journal of Infectious Diseases

journal homepage: www.elsevier.com/locate/ijid

Prevalence of Hepatitis B virus and Hepatitis D virus Coinfection in Western Burkina Faso and molecular characterization of the detected virus strains

Armel M. Sanou[a,*], Kenza Benkirane[b], Bachirou Tinto[c], Assana Cissé[a], Tani Sagna[a], Abdoul Kader Ilboudo[a], Claire Dording[b], Zekiba Tarnagda[a], Claude P. Muller[b,d], Judith M. Hübschen[b]

[a] Unité des Maladies à potentiel épidémiques, Maladies émergentes et Zoonoses, Institut de Recherche en Sciences de la Santé, Bobo-Dioulasso, Burkina Faso
[b] Infectious Diseases Research Unit, Department of Infection and Immunity, Luxembourg Institute of Health, Esch-sur-Alzette, Luxembourg
[c] Département des Sciences Biomédicales, Laboratoire National de Référence des Fièvres Hémorragiques Virales, Centre Muraz, Bobo-Dioulasso, Burkina Faso
[d] Laboratoire national de santé, 1, rue Louis Rech • L-3555 Dudelange, Luxembourg

ARTICLE INFO

Article history:
Received 9 October 2017
Received in revised form 1 February 2018
Accepted 2 February 2018
Corresponding Editor: Eskild Petersen, Aarhus, Denmark

Keywords:
HBV
HDV
Seroprevalence
HBV genotype
Burkina Faso

ABSTRACT

Objectives: In this study, we monitored the seroprevalence of HBV-HDV co-infection in different population groups in the Western part of Burkina Faso, and described the genetic diversity of the detected virus strains.
Methods: Between October 2013 and December 2014, venous blood samples were collected from different cohorts (blood donors, pregnant women, outpatients) in the western region of Burkina Faso. Samples were tested for HBsAg and total anti-HDV antibodies. Positive samples were further analysed for HBV-DNA and HDV-RNA. Genotyping of the detected virus strains was done by nucleotide sequencing and phylogenetic analyses.
Results: A total of 841 participants were included in this study. The mean age was 27.45 years (range: 7–89 years). HBsAg was found in 117 (13.9%) participants. Of the HBsAg positive samples, 4 (3.4%) were positive for total anti-HDV antibodies and negative for HDV RNA. Phylogenetic analyses based on the HBV complete genome (n = 10) and S fragment sequences (n = 35) showed that all strains belonged to genotype E.
Conclusions: Our study showed a high HBsAg prevalence, but a low rate of HDV co-infection in HBsAg carriers from western Burkina Faso. The predominance of HBV genotype E in the country was confirmed. Our findings contribute to a better understanding of the burden of HBV and HDV infection in western Burkina Faso.
© 2018 The Author(s). Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Introduction

With about 240 million chronic carriers worldwide and more than 686 000 deaths per year (WHO, 2014), Hepatitis B virus (HBV) infection remains a serious public health problem, particularly in endemic areas including Southeast Asia and Sub-Saharan Africa (WHO, 2015). To date, eight genotypes named A – H are recognized (Norder et al., 2004) and two additional genotypes, I and J, have been proposed (Tran et al., 2008; Tatematsu et al., 2009). HBV genotypes may have distinct geographical distributions. In Africa, especially genotypes A, D and E are prevalent, with genotype E being dominant throughout West Africa (Hubschen et al., 2008; Cooksley, 2010; Kramvis, 2014; Pourkarim et al., 2014; Kramvis, 2016; Hubschen et al., 2011). Several studies have implicated HBV genotypes in disparate disease progression, clinical outcome, therapeutic response and the degree of protection provided by vaccination (Cooksley, 2010; Kramvis, 2014; Pourkarim et al., 2014; Kramvis, 2016).

* Corresponding author at: Armel Moumouni Sanou, Institut de Recherche en Sciences de la Santé (IRSS), 399 Avenue de la liberté, BP 545, Bobo-Dioulasso, Burkina Faso.
E-mail addresses: armelbf@gmail.com (A.M. Sanou), kenza.benkirane@lih.lu (K. Benkirane), tintobachirou@yahoo.fr (B. Tinto), assanacisse@yahoo.fr (A. Cissé), stanilinda@gmail.com (T. Sagna), ilboudokader@yahoo.fr (A.K. Ilboudo), claire.dording@lih.lu (C. Dording), zekiba@hotmail.com (Z. Tarnagda), claude.muller@lih.lu (C.P. Muller), judith.huebschen@lih.lu (J.M. Hübschen).

https://doi.org/10.1016/j.ijid.2018.02.004
1201-9712/© 2018 The Author(s). Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Hepatitis Delta virus (HDV) is a defective virus that depends on HBV envelope proteins for virion assembly and propagation. Co-infection of HBV and HDV is associated more often with severe liver disease compared to HBV mono-infection (Opaleye et al., 2016). Among the 240 million chronic HBV carriers, about 15–20 million are estimated to be infected with HDV (Wedemeyer and Manns, 2010).

Burkina Faso remains an endemic area for HBV infection with an about 10-30% prevalence of hepatitis B surface antigen (HBsAg) carriers (Collenberg et al., 2006; Simpore et al., 2006; Nagalo et al., 2011; Diendéré et al., 2011; Bado et al., 2013). Despite this situation, little is known about HDV prevalence and molecular characteristics of HBV strains in the country. In this study, we monitored the seroprevalence of HBV-HDV co-infection in different population groups in the Western part of Burkina Faso, and described the genetic diversity of the detected virus strains.

Methods

Study population

A total of 841 venous blood samples collected between October 2013 and December 2014 from consecutive pregnant women (n = 500), blood donors (n = 192) and outpatients of the Provincial Hospital of Orodara who had a venepuncture for unrelated reasons (n = 149) were included in this study. The minimum sample size of 547 participants was calculated based on a proportion of 17.3% of HBsAg carriers among pregnant women/blood donors (Collenberg et al., 2006) using Schwartz's formula. Participants were enrolled after written informed consent in the Healthcare Center of Accart-Ville in Bobo-Dioulasso, the blood transfusion center of Bobo-Dioulasso and the Provincial Hospital of Orodara, respectively. After recording sociodemographic data using a structured questionnaire, blood samples were taken in dry tubes (BD Vacutainer, UK) and the sera were stored at −80 °C until analysis.

Serological tests

Serological tests were performed at the Virology laboratory of the "Institut de Recherche en Sciences de la Santé" (IRSS), Burkina Faso. HBsAg was tested using the Murex HBsAg version 3 assay (Diasorin, Italy) and positive samples were further screened for total HDV antibodies (HDV-Ab) using ETI-AB-DELTAK-2 kits (Diasorin, Italy). All tests were performed and interpreted in accordance with the manufacturers' instructions.

HBV and HDV detection and molecular characterization of HBV strains

Viral DNA was purified from 200 µl of serum using QIAamp DNA Blood Mini Kits (Qiagen, Germany). Since according to the manufacturer these kits co-purify RNA, though with suboptimal yield, the same material was used for the detection of HDV RNA. The extract was stored at −80 °C until shipment on dry ice to Luxembourg for molecular investigation. Real-time PCRs were performed to detect HBV DNA in HBsAg positive samples (Liu et al., 2007) and HDV RNA in HDV-Ab positive samples (Le Gal et al., 2005). To characterize the HBV strains, the S gene region or the complete genome were amplified as described previously (Olinger et al., 2006). PCR products were purified using the JetQuick PCR Purification Spin kit (Genomed, Germany) or the QIAquick Gel extraction kit (Qiagen). Sequencing was performed with the semi-nested primers used for amplification, the Big Dye Terminator version 3.1 cycle sequencing kit (Life Technologies, Belgium) and the ABI 3130 Avant capillary sequencer (Applied Biosystems, US). Sequences were edited with SeqScape v2.5 (Applied Biosystems) and aligned with reference sequences using BioEdit version 7.2.5 (Hall, 1999). Phylogenetic analyses based on the Kimura 2-parameter model and the Neighbor-Joining algorithm were conducted with MEGA version 6 (Tamura et al., 2013). Sequences obtained in this study are available under DDBJ/EMBL/GenBank accession numbers **LT623842-LT623886**.

Statistical analysis

Data were entered and checked for accuracy using Epidata software, version 3.1. Analyses were carried out using STATA software version 14.2. Chi-square and Fisher's exact tests were used to assess associations between categorical variables. P-values below 0.05 were considered statistically significant.

Results

Study population characteristics

A total of 841 participants aged between 7 and 89 years (mean age 27.45 ± 9.77 years) were included in this study and the male/female ratio was 0.37. Five hundred eighty-eight participants (69.9%) were married and more than one third (34.1%) never visited a school (Table 1). Twelve (1.4%) participants were vaccinated against HBV, 546 (64.9%) had no information about their immunization status and all others reported to be unvaccinated. Potential risk factors for HBV/HDV infection included injection, excision, dental care, scarification, surgery and blood transfusion (Table 1).

HBV and HDV seroprevalence and associated risk factors

The overall HBsAg prevalence was 13.9% (117/841) and the cohort-specific rates were 10.7% among outpatients, 13.0% among blood donors and 15.2% among pregnant women (p = 0.355). Differences in HBsAg positivity were not related to gender (p = 0.604), marital status (p = 0.966) or level of education (p = 0.967), but a negative correlation was found for dental care (17 HBsAg positives among 195 people with dental care (8.7%) versus 100 HBsAg positives among 646 without (15.5%), p = 0.017). All three blood donors with a history of surgery were also HBsAg positive (p = 0.002), while there was no positive correlation for the other cohorts or overall. HBsAg prevalence was higher among participants up to 35 years of age (14.9%; 109/731) than among participants of at least 36 years (7.3%; 8/110; p = 0.764), with no significant differences between the cohorts (p = 0.097). The HBsAg positive participants were either unvaccinated (29.9%; 35/117) or had an unknown vaccination status (70.1%; 82/117), but no statistically significant correlation was found (p = 0.442). Total anti-HDV antibodies were detected in 4 of 117 (3.4%) HBsAg carriers. Three of the anti-HDV antibody positive participants were pregnant women and one was a male blood donor (p = 0.643). The HDV-Ab positive patients were between 24 and 26 years old and 3 of them were married. Three had an unknown vaccination status and all four levels of education were represented. Concerning potential risk factors, two HDV-Ab positive participants each reported scarification, excision and injection.

Molecular characterization of detected virus strains

HBV DNA was detected in 50 of the 117 HBsAg positive specimens (42.7%). The complete genome was amplified for 10 samples (10/50; 20%) and a 735 bp region comprising the complete S gene was obtained for 35additional samples (35/50; 70%), while sequences of the remaining 5 PCR positives were of insufficient quality. The phylogenetic analyses of the complete genome sequences (Figure 1a) and the S region (Figure 1b) showed that

Table 1
HBsAg positivity in relation to sociodemographic characteristics and potential risk factors.

	HBsAg positives among			
	Pregnant women n = 500	Blood donors n = 192	Outpatients n = 149	Total cohort n = 841
Age groups (years)				
≤25	43/279 (15.4%)	16/120 (13.3%)	6/40 (15.0%)	65/439 (14.8%)
≥26 – 35	32/195 (16.4%)	8/63 (12.7%)	4/34 (11.8%)	44/292 (15.1%)
≥36	1/26 (3.8%)	1/9 (11.1%)	6/75 (8.0%)	8/110 (7.3%)
Gender				
Male	–	23/154 (14.9%)	6/71 (8.5%)	29/225 (12.9%)
Female	76/500 (15.2%)	2/38 (5.3%)	10/78 (12.8%)	88/616 (14.3%)
Marital status				
Single	10/66 (15.2%)	22/160 (13.8%)	3/27 (11.1%)	35/253 (13.8%)
Married	66/434 (15.2%)	3/32 (9.4%)	13/122 (10.7%)	82/588 (13.9%)
Level of education				
Unschooled	34/234 (14.5%)	0/3 (0.0%)	8/50 (16.0%)	42/287 (14.6%)
Elementary school	22/142 (15.5%)	2/10 (20.0%)	1/35 (2.9%)	25/187 (13.4%)
Secondary school	19/116 (16.4%)	9/76 (11.8%)	5/56 (8.9%)	32/248 (12.9%)
University	1/8 (12.5%)	14/103 (13.6%)	2/8 (25.0%)	17/119 (14.3%)
Vaccination status				
Vaccinated	1/500 (0.2%)	4/192 (2.1%)	7/149 (4.7%)	12/841 (1.4%)
Unvaccinated	38/500 (7.6%)	172/192 (89.6%)	73/149 (49.0%)	283/841 (33.7%)
Unknown	461/500 (92.2%)	16/192 (8.3%)	69/149 (46.3%)	546/841 (64.9%)
Potential risk factors				
Blood transfusion	1/14 (7.1%)	0/1 (0.0%)	0/5 (0.0%)	1/20 (5.0%)
Surgery	4/19 (21.1%)	3/3 (100.0%)	2/25 (8.0%)	9/47 (19.1%)
Scarification	27/141 (19.1%)	0/4 (0.0%)	2/9 (22.2%)	29/154 (18.8%)
Dental care	13/116 (11.2%)	2/26 (7.7%)	2/53 (3.8%)	17/195 (8.7%)
Excision	51/365 (14.0%)	1/10 (10.0%)	8/53 (15.1%)	60/428 (14.0%)
Injection	68/425 (16.0%)	0/13 (0.0%)	12/120 (10.0%)	80/558 (14.3%)

all HBV strains belonged to genotype E. Sequences from Guinea and Saudi Arabia from between 2006 and 2013 were the closest BLAST fits of the complete genome sequences from Burkina (besides one sequence with unknown origin named AB320, Figure 1a). The overall mean distance among the complete genome sequences from Burkina was 1.87%, with a minimum and maximum distance of 1.14 and 2.72%, respectively. For the S fragment sequences, the corresponding numbers were 1.00%, 0 and 3.22%. HDV RNA was not detected in any of the four antibody positive patients.

Discussion

This study confirmed the high prevalence of HBV infection in Burkina Faso, with an overall HBsAg positivity of 13.9%. Together with previous studies reporting prevalence rates between 9.8% and 27.3% (Collenberg et al., 2006; Simpore et al., 2006; Sangaré et al., 2009; Nagalo et al., 2012; Candotti et al., 2016), this shows that HBV infection is a major public health problem. A more systematic monitoring of HBV infection (and HDV co-infection) would allow to adapt treatment schemes. Since HBV vaccine protects against both HBV and HDV infection (also in our study none of the HBsAg positive participants had evidence of previous HBV vaccination), routine childhood vaccination against HBV should be strengthened in Burkina.

Despite the high burden of HDV infection among chronic carriers of HBsAg worldwide (Abbas et al., 2010; Yacoubi et al., 2015), little is known about the seroprevalence of HDV antibodies in Burkina Faso. Our study showed that 3.4% (4/117) of HBsAg carriers were positive for total HDV-Ab. Other studies conducted in Bobo-Dioulasso reported prevalences of 3.38% and 2.50% among blood donors and a cohort of mothers, respectively (Andernach et al., 2014; Sawadogo et al., 2015). These results suggest that Burkina Faso is a low endemic area for HDV infection. Nevertheless, a large-scale study is needed to confirm these findings, especially since higher prevalence rates were reported in a cohort of children (9/44, 20.5%) (Andernach et al., 2014), in a cohort of HIV patients in West Africa including Burkina Faso (10/67, 14.9%) (Coffie et al., 2017) and in neighbouring countries (De Paschale et al., 2014; Asmah et al., 2014).

In contrast to previous studies (Komas et al., 2010; Dashtseren et al., 2017), we identified a negative correlation between dental treatment and risk of acquiring HBV infection. Preventive procedures implemented by dentists due to HIV and other blood-borne viral infections certainly play a role for the decreased risk of infection (Jakubovics et al., 2014). Whether people who seek dental care are in general more concerned about their health and thus more likely to avoid HBV exposure, deserves further study.

Interestingly, our study showed a higher prevalence of HBsAg in participants younger than 36 years of age in each cohort and overall as compared to older participants. The prevalence in the latter population group may be somewhat underestimated due to the rather low number of participants (n = 110; 13.1%) belonging mostly to the outpatient cohort, but may also be due to a less risky sexual behaviour than in the younger age groups. Previous studies characterized HBV strains from Burkina Faso as belonging to genotype E and quasisubgenotype A3 (Hubschen et al., 2008; Candotti et al., 2016; Compaore et al. 2016). Our data confirm the predominance of genotype E in the country and the region (Hubschen et al., 2008; Candotti et al., 2016; Traoré et al., 2015; Forbi et al., 2013). No genotype A strains were found, possibly because this genotype shows a restricted geographical distribution

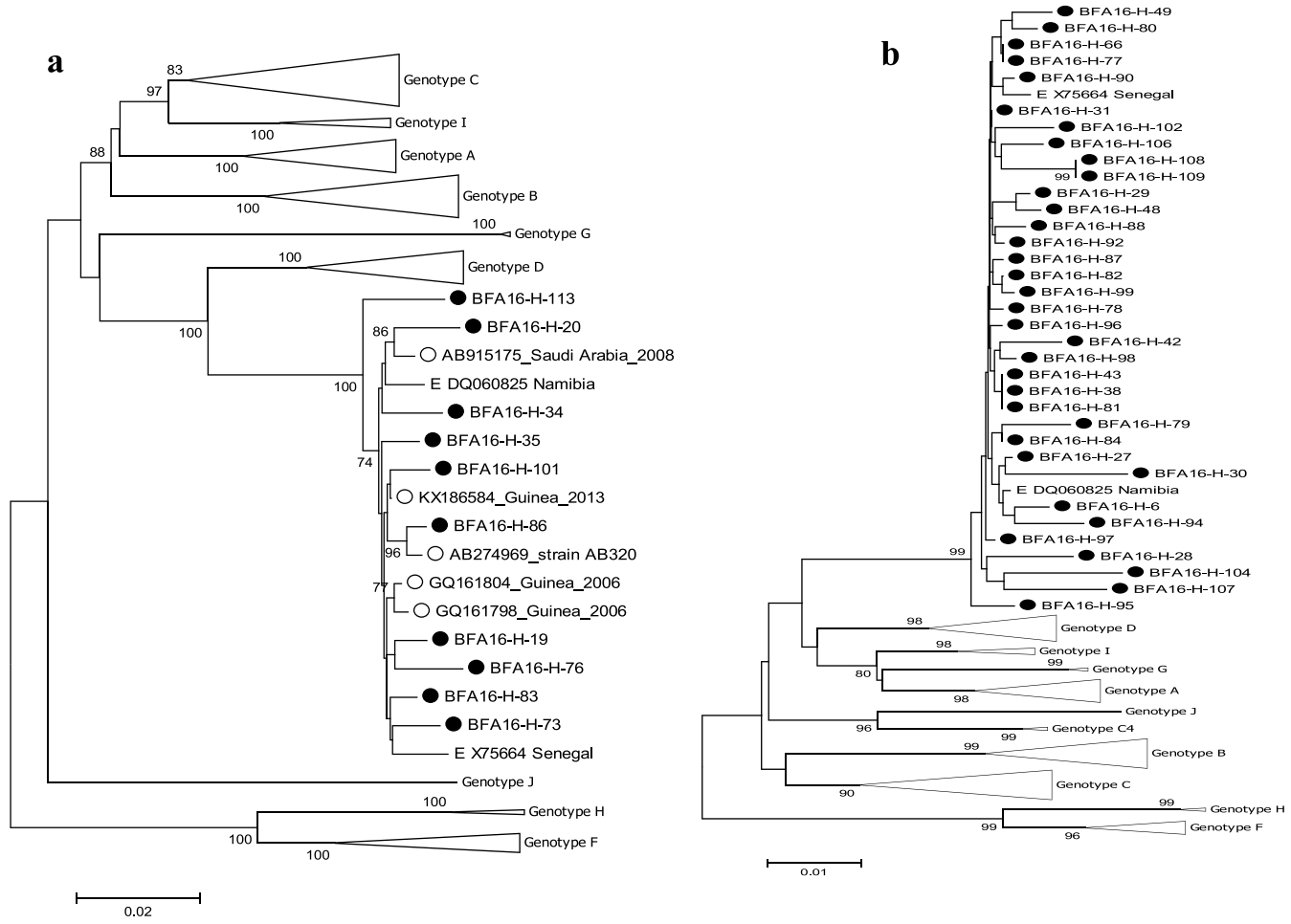


Figure 1. (a) Phylogenetic tree of complete genome sequences of HBV constructed using the Neighbour-Joining algorithm and MEGA 6.0 software. HBV sequences from this study are marked by the symbol • and the closest BLAST fits by ○. Bootstrap values ≥70 (1000 replicates) are shown.
(b) Phylogenetic tree based on 735 nucleotides of the S fragment constructed using the Neighbour-Joining algorithm and MEGA 6.0 software. HBV sequences from this study are marked by the symbol •. Bootstrap values ≥70 (1000 replicates) are shown.

in the country, since it was so far reported only from the central part (Candotti et al., 2016). The analysis of S fragment sequences showed a mean genetic distance of 1.00% (range: 0.00–3.22%). No epidemiological link other than that they originated from the same locations, respectively, was found for the patients with identical sequences (66 and 77 were from Bobo-Dioulasso, 108 and 109 were from Orodara; 38, 43 and 81 were from Bobo-Dioulasso). The genetic distances were comparable to the previously reported 1.45% (range: 0.00–5.93%) among blood donors (Candotti et al., 2016) and further corroborate the hypothesis that genotype E has a rather short evolutionary history (Mulders et al., 2004; Andernach et al., 2009). Interestingly, the HBV genotype E complete genome sequences from Burkina Faso were closely related to sequences from Guinea and Saudi Arabia, with which there are contacts in the context of e.g. trade and emigration (Bouda, 2009).

In this study, only 50 out of 117 HBsAg positive samples were HBV DNA positive, suggesting that a low viral load, DNA denaturation during sample storage and/or a poor quality of the extracted DNA may have played a role. Mismatches at the primer binding sites that could have prevented amplification were not identified for genotype E strains. HDV RNA was not detected in any of the four antibody positive participants, maybe because no chronic infection was established, but possibly also because of suboptimal RNA yield obtained with the DNA extraction kits.

Conclusion

We found a high HBsAg prevalence and a low prevalence of HDV antibodies in HBsAg carriers in the Western region of Burkina Faso. Our study provided the first HBV genotype information from that part of the country, confirming the predominance of genotype E in Burkina and in West Africa. The results contribute to a better understanding of the burden of HBV and HDV infection in western Burkina Faso and of the circulating HBV strains.

Funding

The present study received financial support from the Luxembourg Ministry of Foreign and European Affairs and the “Institut de Recherche en Sciences de la Santé” in Burkina Faso.

Authors’ contributions

AMS, ZT and JMH developed the research idea and were involved in the study design, the data analysis and the drafting of the manuscript. CPM critically reviewed the article. AMS, BT and AKI collected the sera and compiled the epidemiological data. AMS, BT, AC and TS performed the serological analysis and the extraction of viral nucleic acids. KB and CD did the HBV S fragment and

complete genome amplification PCRs and the sequencing. JMH performed the sequence and phylogenetic analyses. All authors read and approved the final version of the manuscript.

Conflicts of interest

None

Ethical approval

This study was approved by the Institutional Ethical Committee of the Centre Muraz (N° A20-2011/CE-CM) according to the declaration of Helsinki. Informed consent was obtained from all participants or their parents before enrolment in the study. The participants tested positive for HBsAg were referred to the Teaching hospital Souro Sanou of Bobo-Dioulasso for further investigation and treatment.

Acknowledgments

We thank all healthcare workers of the different centres for their technical assistance during this study. We also thank the regional director of health for his support during the administrative procedure.

References

Abbas Z, Jafri W, Raza S, Hepatitis D. Scenario in the Asia-Pacific region. World J Gastroenterol 2010;16(5):554–62.
Andernach IE, Judith MH, Muller CP. Hepatitis B virus: the genotype E puzzle. Rev Med Virol 2009;19:231–40.
Andernach IE, Leiss KV, Tarnagda Z, Tahita MC, Otegbayo JA, Forbi JC, et al. Characterization of hepatitis delta virus in sub-Saharan Africa. J Clin Microbiol 2014;52(5):1629–36.
Asmah RH, Boamah I, Afodzinu M, Brown CA, Brandful J, Adjei DN, et al. Prevalence of hepatitis d infection in patients with hepatitis B virus-related liver diseases in Accra, Ghana. West Afr J Med 2014;33(1):32–6.
Bado G, Penot P, N'Diaye MD, Amiel C, Hema A, Kamboule EB, et al. Hepatitis B seroprevalence in HIV-infected patients consulting in a public day care unit in Bobo Dioulasso, Burkina Faso. Med Mal Infect 2013;43(5):202–7.
Bouda E. Le rattachement des burkinabé de l'étranger à leur pays d'origine et leur apport au développement. Memoire Online, Economie et Finance. 2009 http://www.memoireonline.com/03/10/3237/m_Le-rattachement-des-burkinabe-de-letranger?leur-pays-dorigine-et-leur-apport-au-developp1.html. [Assessed 28 March 2017].
Candotti D, Diarra B, Bisseye C, Tao I, Quang KP, Sanou M, et al. Molecular characterization of hepatitis B virus in blood donors from Burkina Faso: prevalence of quasi-subgenotype A3, genotype E, and mixed infections. J Med Virol 2016;88(12):2145–56.
Coffie PA, Tchounga BK, Bado G, Kabran M, Minta DK, Wandeler G, et al. Prevalence of hepatitis B and delta according to HIV-type: a multi-country cross-sectional survey in West Africa. BMC Infect Dis 2017;17(1):466–74.
Collenberg E, Ouedraogo T, Ganame J, Fickenscher H, Kynast-Wolf G, Becher H, et al. Seroprevalence of six different viruses among pregnant women and blood donors in rural and urban Burkina Faso: a comparative analysis. J Med Virol 2006;78(5):683–92.
Compaore TR, Diarra B, Assih M, Obiri-Yeboah D, Soubeiga ST, Ouattara AK, et al. HBV/HIV co-infection and APOBEC3G polymorphisms in a population from Burkina Faso. BMC Infect Dis 2016;16:336–43.
Cooksley WG. Do we need to determine viral genotype in treating chronic hepatitis B?. J Viral Hepat 2010;17(9):601–10.
Dashtseren B, Bungert A, Bat-Ulzii P, Enkhbat M, Lkhagva-Ochir O, Jargalsaikhan G, et al. Endemic prevalence of hepatitis B and C in Mongolia: a nationwide survey among Mongolian adults. J Viral Hepat 2017;17(10):601–10.
De Paschale M, Ceriani C, Cerulli T, Cagnin D, Cavallari S, Ndayake J, et al. Prevalence of HBV, HDV, HCV, and HIV infection during pregnancy in northern Benin. J Med Virol 2014;86(8):1281–7.
Diendéré EA, Tiéno H, Bognounou R, Ouédraogo DD, Simporé J, Ouédraogo/Traoré R, et al. Prévalences et facteurs associés aux infections à Virus de l'Immunodéficience Humaine et virus de l'hépatite B, à la syphilis et à la tuberculose pulmonaire bacillifère en milieu carcéral au Burkina Faso. Med Trop 2011;71:464–7.
Forbi JC, Ben-Ayed Y, Xia GL, Vaughan G, Drobeniuc J, Switzer WM, et al. Disparate distribution of hepatitis B virus genotypes in four subSaharan African countries. J Clin Virol 2013;58(1):59–66, doi:http://dx.doi.org/10.1016/j.jcv.2013.06.028.
Hall TA. BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. Nucl Acids Symp Ser 1999;41:95–8.
Hubschen JM, Andernach IE, Muller CP. Hepatitis B virus genotype E variability in Africa. J Clin Virol 2008;43(4):376–80.
Hubschen JM, Mbah PO, Forbi JC, Otegbayo JA, Olinger CM, Charpentier E, et al. Detection of a new subgenotype of hepatitis B virus genotype A in Cameroon but not in neighbouring Nigeria. Clin Microbiol Infect 2011;17:88–94.
Jakubovics N, Greenwood M, Meechan JG. General medicine and surgery for dental practitioners: part 4. Infections and infection control. Br Dent J 2014;217(2):73–7.
Komas NP, Baï-Sepou S, Manirakiza A, Léal J, Béré A, Le Faou A. The prevalence of hepatitis B virus markers in acohort of students in Bangui, Central African Republic. BMC Infect Dis 2010;10:226–32.
Kramvis A. Genotypes and genetic variability of hepatitis B virus. Intervirology 2014;57(3–4):141–50.
Kramvis A. The clinical implications of hepatitis B virus genotypes and HBeAg in pediatrics. Rev Med Virol 2016;26(4):285–303.
Le Gal F, Gordien E, Affolabi D, Hanslik T, Alloui C, Dény P, et al. Quantification of hepatitis delta virus RNA in serum by consensus real-time PCR indicates different patterns of virological response to interferon therapy in chronically infected patients. J Clin Microbiol 2005;43(5):2363–9.
Liu Y, Hussain M, Wong S, Fung SK, Yim HJ, Lok AS. A genotype-independent real-time PCR assay for quantification of hepatitis B virus DNA. J Clin Microbiol 2007;45(2):553–8.
Mulders NM, Venard V, Njayou M, Edorh AP, Oyefolu AOB, Kehinde MO, et al. Low Genetic Diversity despite Hyperendemicity of Hepatitis B Virus Genotype E throughout West Africa. J Infect Dis 2004;190:400–8.
Nagalo BM, Bisseye C, Sanou M, Kienou K, Nebie YK, Kiba A, et al. Seroprevalence and incidence of transfusion-transmitted infectious diseases among blood donors from regional blood transfusion centres in Burkina Faso, West Africa. Trop Med Int Health 2012;17(2):247–53.
Nagalo MB, Sanou M, Bisseye C, Kaboré MI, Nebie YK, Kienou K, et al. Seroprevalence of human immunodeficiency virus, hepatitis B and C viruses and syphilis among blood donors in Koudougou (Burkina Faso) in 2009. Blood Transfus 2011;9(4):419–24.
Norder H, Couroucée AM, Coursaget P, Echevarria JM, Lee SD, Mushahwar IK, et al. Genetic diversity of hepatitis B virus strains derived worldwide: genotypes, subgenotypes, and HBsAg subtypes. Intervirology 2004;47(6):289–309.
Olinger CM, Venard V, Njayou M, Oyefolu AOB, Maïga I, Kemp AJ, et al. Phylogenetic analysis of the precore/core gene of hepatitis B virus genotypes E and A in West Africa: new subtypes, mixed infections and recombinations. J Gen Virol 2006;87(Pt 5):1163–73.
Opaleye OO, Japhet OM, Adewumi OM, Omoruyi EC, Akanbi OA, Oluremi AS, et al. Molecular epidemiology of hepatitis D virus circulating in Southwestern Nigeria. Virol J 2016;13:61–9.
Pourkarim MR, Amini-Bavil-Olyaee S, Kurbanov F, Ranst VM, Tacke F. Molecular identification of hepatitis B virus genotypes/subgenotypes: revised classification hurdles and updated resolutions. World J Gastroenterol 2014;20(23):7152–68.
Sangaré L, Sombié R, Combasséré AW, Kouanda A, Kania D, Zerbo O, et al. Transmission anténatale du virus de l'hépatite B en zone de prévalence modérée du VIH, Ouagadougou, Burkina Faso. Bull Soc Pathol Exot 2009;102(4):226–9.
Sawadogo A, Ouédraogo AS, Poda A, Dahourou H, Pivert A, Mansour W, et al. Séroprévalence de l'infection par le virus de l'hépatite D dans une population de donneurs de sang porteurs de l'Antigène HBs au Centre régional de transfusion sanguine de Bobo-Dioulasso. J Afr Hépatol Gastroentérol 2015;10(1):31–3.
Simpore J, Savadogo A, Ilboudo D, Nadambega MC, Esposito M, Yara J, et al. Toxoplasma gondii, HCV, and HBV seroprevalence and co-infection among HIV-positive and -negative pregnant women in Burkina Faso. J Med Virol 2006;78(6):730–3.
Tamura K, Stecher G, Peterson D, Filipski A, Kuma S. MEGA6: molecular evolutionary genetics analysis version 6.0. Mol Biol Evol 2013;30(12):2725–9.
Tatematsu K, Tanaka Y, Kurbanov F, Sugauchi F, Mano S, Maeshiro T, et al. A genetic variant of hepatitis B virus divergent from known human and ape genotypes isolated from a Japanese patient and provisionally assigned to new genotype J. J Virol 2009;83(20):10538–47.
Tran TT, Trinh TN, Abe K. New complex recombinant genotype of hepatitis B virus identified in Vietnam. J Virol 2008;82(11):5657–63.
Traoré F, Gormally E, Villar S, Friesen MD, Groopman JD, Vernet G, et al. Molecular characteristics of Hepatitis B and chronic liver disease in a cohort of HB carriers from Bamako, Mali. BMC Infect Dis 2015;15:180–8.
Wedemeyer H, Manns MP. Epidemiology, pathogenesis and management of hepatitis D: update and challenges ahead. Nat Rev Gastroenterol Hepatol 2010;7(1):31–40.
World Health Organization (WHO). Hepatitis B Fact sheet no 204. 2014 http://www.who.int/mediacentre/factsheets/fs204/en/. [Assessed 15 October 2016].
World Health Organization (WHO). Guidelines for the prevention, care and treatment of persons with chronic hepatitis B infection. Geneva: WHO; 2015 http://www.who.int/hiv/pub/hepatitis/hepatitis-b-guidelines/en/. [Assessed 15 October 2016].
Yacoubi L, Brichler S, Mansour W, Le Gal F, Hammami W, Sadraou A, et al. Molecular epidemiology of hepatitis B and Delta virus strains that spread in the Mediterranean North East Coast of Tunisia. J Clin Virol 2015;72:126–32.